Research Paradigm Justification Statement (revised v1.1)

PhD in Deep Tech — UNMSM · Session 1 · Juseph Montesinos Asmat

1. Topic and context

The world's data centers now consume more than 130 GW of electricity, and Gartner projects demand to triple by 2030, driven by artificial intelligence, amid mounting pressure on energy and water. Peru, with only 14 data centers, lags its regional peers. INSUR, an insurer supervised by the SBS and subject to Law N° 29733, operates a small, standards-compliant data center whose components are individually manageable — servers, storage, and network with telemetry via XClarity, Spectrum Control, and VMware — yet without a DCIM or comprehensive instrumentation. This is the prevailing reality of mid-sized Peruvian data centers; existing digital twins assume full telemetry, opening a design gap relevant to Law N° 31814 and its Regulation (DS 115-2025-PCM).

2. Research question

How can an *advisory* (non-closed-loop) digital twin for decision support and energy-and-capacity optimization be designed for a Peruvian data center with limited telemetry (the INSUR case), and how can it be evaluated — both its validity as a model and its utility for decisions — as improving operational decisions over manual management, such that its design principles are transferable to other sites?

3. Paradigm, epistemology, and justification

Epistemologically, I adopt **pragmatism**: claims are justified by their demonstrated usefulness in action, not by correspondence to an objective reality or by intersubjective meaning. The chosen paradigm is therefore **Design Science**, whose purpose is to produce artifacts evaluated by their utility (Hevner et al., 2004). The quantitative evaluation of my artifact does not compete with this paradigm but belongs to its evaluative repertoire; I do not adopt positivism as the governing paradigm, but rather borrow its measurement techniques inside the design cycle. I reject interpretivism because the core of the problem is infrastructure performance, not the subjective meaning of human experience. I reject mixed methods because, although validation will use quantitative metrics, the central contribution is an artifact and its design principles, not a quali-quanti methodological integration.

4. Implications and evaluation design

- **Methodology:** relevance, rigor, and design cycles; build and evaluate the artifact.
- **Artifact:** an advisory planning-and-optimization digital twin that synthesizes available component-level telemetry (power and utilization via XClarity/VMware, storage via Spectrum Control, power via managed PDUs and in-row cooling) into a unified DCIM-like view.
- **Model validity (before utility):** validate the twin's predictions against held-out historical periods and report estimation uncertainty — critical given sparse telemetry.
- **Decision utility:** compare assisted decisions against manual management using two metrics **normalized for ambient temperature and IT load**, with a baseline defined in advance and interrupted-time-series logic: (a) **PUE** (infrastructure efficiency) and (b) **energy proportionality** (useful compute work per unit of energy), capturing the IT-side efficiency that PUE ignores.
- **Generalization (FEDS):** naturalistic summative evaluation at INSUR, with formative episodes during development, and re-validation at a second site (Policlínico Peruano Japonés) to abstract transferable design principles (Venable et al., 2016).

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- **Data governance:** data access and confidentiality agreements with INSUR, anonymization, and compliance with Law N° 29733 and SBS supervision, given the handling of operational data from a regulated insurer.
- **Target venue:** DESRIST (the reference DSR community), with a pathway to Energy Informatics and Sustainable Computing.

5. Doubt or tension

The main tension is whether, without additional instrumentation or real-time telemetry, a twin built on sparse component-level telemetry can achieve model validity and decision improvements measurable enough to justify, in the future, the investment in additional sensors and actuators.

Academic references

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Peruvian legal norms

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Data and industry sources

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